

GalaxySagittarius: Structure- and Similarity-Based Prediction of Protein Targets for Druglike Compounds

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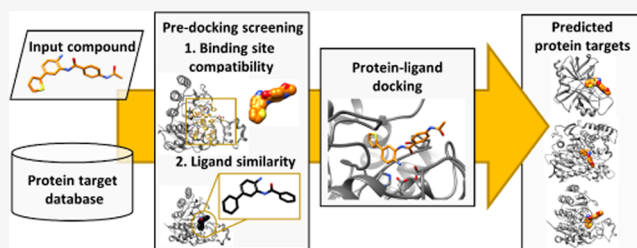


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ABSTRACT: Computational techniques for predicting interactions of proteins and druglike molecules have often been used to search for compounds that bind a given protein with high affinity. More recently, such tools have also been applied to the reverse procedure of searching protein targets for a given compound. Among methods for predicting protein–ligand interactions, ligand-based methods relying on similarity to ligands of known interactions are effective only when similar protein–ligand interactions are known. Receptor-based methods predicting protein–ligand interactions by molecular docking are effective only when high-accuracy receptor structures and binding sites are available. Moreover, the computational cost of molecular docking tends to be too high to be applied to the entire protein structure database. In this paper, an effective target prediction method, which combines ligand similarity-based and receptor structure-based approaches, is introduced. In this method, protein–ligand docking is performed after efficient structure- and similarity-based screening. The enriched protein target database by predicted binding ligands and sites allows detection of protein targets with previously unknown ligand interactions. The method, called GalaxySagittarius, is freely available as a web server at <http://galaxy.seoklab.org/sagittarius>.



1. INTRODUCTION

Traditional computational studies of protein–ligand interactions involve calculating binding affinities and/or binding poses to find high-affinity ligands for a given protein target.^{1–4} Recently, method developments for predicting protein targets interacting with a given ligand have become more active, with increasingly more sophisticated computational techniques including machine learning^{5,6} and advanced computer hardware technologies. Such target prediction methods can potentially play important roles in biochemical studies of various chemicals involved in multiple protein interaction networks, studies of possible side effects of druglike molecules, and drug repositioning. Drug repositioning, a strategy of applying an existing drug to treat a new disease indication,^{7–10} has been attracting great interests owing to the reduced cost and time for drug development.^{11,12}

There are two major types of computational target prediction methods, ligand-based and receptor-based methods. Ligand-based methods are based on the belief that proteins can be targeted similarly by similar compounds. Based on the accumulated knowledge on drug–target interactions, similarities of a query compound to the compounds known to interact with candidate proteins are evaluated to find new protein targets for the query compound. Various methods based on ligand similarity have been developed, including ChemMapper,¹³ ChemProt 3.0,¹⁴ and SwissTargetPrediction.¹⁵ Receptor-

based target prediction, also known as inverse screening, employs molecular docking methods to search for protein binding sites suitable for a query molecule. In contrast to virtual screening, inverse screening performs docking of a ligand to multiple protein binding sites to identify protein–ligand interactions. Several protein–ligand docking methods have been adopted for target prediction purposes, resulting in efficient inverse screening tools such as TarFisDock,¹⁶ idTarget,¹⁷ and ACID.¹⁸

Although both types of target prediction methods have shown promising results, there are still clear limitations in the currently available methods. In the case of ligand-based methods, target search is restricted to proteins known to have active interactions with certain compounds, so the outcome of the search relies highly on available activity data. Receptor-based methods do not require known active compounds for proteins but do require accurate protein structures and binding sites for docking. Moreover, docking tends to be much more computationally expensive than ligand-

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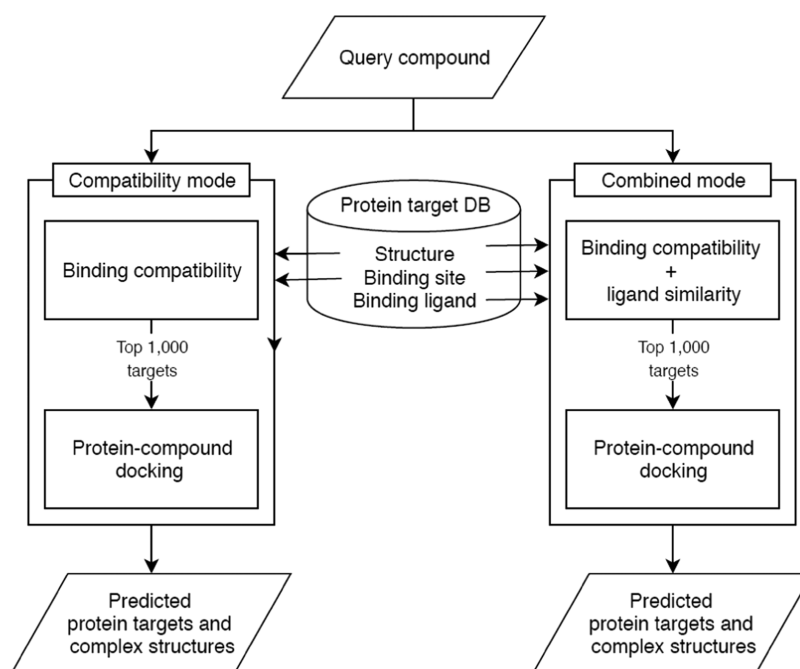


Figure 1. Overall target prediction procedure of GalaxySagittarius.

based methods, making it hard to be applied to a large protein structure database.

In this paper, a new target prediction method, which combines benefits of both ligand-based and receptor-based methods, is introduced. Receptor structure-based and ligand similarity-based fast screening methods are applied first on the entire database to select candidate targets, and more time-consuming docking calculations are performed on the selected candidates. For structure-based screening, a machine learning method that evaluates physicochemical compatibility between a query molecule and a binding site was developed. The structure database of human proteins constructed for this method consists not only of experimentally resolved protein structures and protein–ligand interactions but also of predicted interactions. The predicted binding ligands and sites greatly enrich the target database. Such added information had positive impacts on both structure- and similarity-based components of the target prediction protocols in test cases.

This program, called GalaxySagittarius, is available as a free web-based platform with a user-friendly interface to aid nonexpert users in target prediction tasks.

2. METHODS

2.1. Overall Procedure. The overall target prediction protocol for GalaxySagittarius is shown in Figure 1. Given a query compound, protein targets and target–compound complex structures are predicted in two parallel modes. The two modes are different only in the predocking stage that screens the large target database before computationally expensive docking. The first prediction mode, called the compatibility mode, employs a random forest model to predict the compatibility of the query compound to protein binding sites for predocking screening. This screening is performed against the entire protein target database. The second prediction mode, called the combined mode, exploits the similarity of the query compound to known or predicted ligands of proteins in addition to the predicted binding site

compatibility. Only the protein targets with available ligand information (either known or predicted) are considered in this mode. While the combined mode is expected to be effective in finding targets that are known to bind compounds similar to the query compound, the compatibility mode is expected to be more effective in finding novel targets for which no ligand information is available.

The query compound is docked to each of the 1000 proteins selected in the predocking stage of each prediction mode using a protein–ligand docking program GalaxyDock2¹⁹ to rerank the target candidates and to obtain the target–compound complex structures. The ranked proteins and complex structures can be downloaded from the web server, and the top 200 proteins can be observed on the server.

2.2. Protein Target Database. The protein target database consists of experimentally resolved structures of human proteins deposited in the protein data bank (PDB) and ligand binding sites and corresponding ligands extracted from PDB and predicted by GalaxySite²⁰ and P2Rank.²¹ Protein structures without side-chain coordinates were filtered out. Only druglike small molecules were included as ligands, excluding solvent molecules, ions, and other small molecules with less than five non-hydrogen heavy atoms. In the target database, kinases are the most abundant protein family covering 6.84% of the database, followed by proteases (4.31%), GPCRs (1.49%), and ion channels (1.36%).

GalaxySite predicts both sites and ligands, and P2Rank predicts sites only. Protein structures and binding sites are used for docking as well as for binding site compatibility prediction, and binding ligands are used for ligand similarity-based screening. Multiple structures for the same protein chains were kept to account for structural differences arising from protein flexibility. Binding sites located at protein interfaces were also kept to account for ligands interacting with multiple human protein chains simultaneously.

A similarity-based binding site prediction program GalaxySite²⁰ was used to enrich the database with more potential binding sites and ligands for proteins both with and without

Table 1. Numbers of Different Entries in the GalaxySagittarius Protein Target Database

	with ligand interactions from PDB	with additional predicted ligand interactions by GalaxySite	full DB (with additional predictions by P2Rank)
no. of protein chains	2449	4454	6037
no. of protein structures	20 477	33 381	41 333
no. of protein–ligand interactions (no. of ligands)	45 416 (12 630)	389 237 (13 637)	389 237 (13 637)
no. of binding sites (no. of clusters)	44 805 (7657)	113 172 (18 396)	134 010 (31 552)
no. of protein chains + no. of protein interfaces with binding sites	3779	8279	11 768

bound ligands available in PDB. This method extracts binding sites and ligands of homologous proteins in PDB and carries out molecular docking of the ligands of homologous proteins to the binding sites mapped to the target protein to evaluate the binding prediction. GalaxySite was successfully tested in the CASP blind prediction.²² In this work, the default option was used to run GalaxySite.

For those protein structures without any binding sites from PDB or from GalaxySite, up to three binding sites were predicted by P2Rank²¹ with default options to further enrich the database. P2Rank predicts binding sites using the local chemical environment of the protein surface by machine learning. Binding sites predicted by P2Rank are used only in the compatibility mode of GalaxySagittarius since binding ligands are not predicted.

A large fraction of binding sites in the target database overlaps with each other since they come from multiple structures and/or multiple binding sites for the same protein chain. For the time-consuming docking calculation, only nonredundant binding sites are considered. Redundancy in binding sites was removed by hierarchical clustering after structure superposition of protein chains near the binding site. The number of clusters was determined by setting the maximum distance between two binding site geometric centers at 4.5 Å.

The numbers of different entries in the GalaxySagittarius protein target database are listed in Table 1.

2.3. Predocking Methods for Screening Targets. The predocking screening step aims at fast and accurate search through the protein target database to select proteins to be subjected to the time-consuming docking calculations. Each of the two prediction modes selects 1000 targets. The number 1000 was chosen as a compromise between predictability and run time after trying different numbers.

In the compatibility mode, the compatibility score of the query compound is evaluated for all binding sites (134 010 sites from 11 768 proteins and interfaces according to Table 1). This is the only predocking method applied to the binding sites with no corresponding ligands in PDB or predicted by GalaxySite (20 838 sites from 3489 proteins and interfaces). In the combined mode, the predocking score combining ligand similarity is evaluated for all binding sites with available ligands (113 172 sites from 8279 proteins and interfaces).

The number of binding sites is much larger than that of proteins because multiple binding sites arise from multiple protein structures, different binding pockets, and/or various protein–protein interfaces. When selecting 1000 targets and the corresponding binding sites for docking from 11 768 (8279) proteins and protein interfaces in the compatibility (combined) mode, only the best-scoring binding site for each protein/interface is considered.

The predocking score of each mode is a probability score for the query compound to be an actual ligand at a given binding

site. Compound–site pairs for training were derived from (1) PDBBind²³ for the compatibility mode because this set consists of proteins with experimentally verified binding sites that can be used for testing binding site compatibility prediction and (2) ChEMBL25²⁴ for the combined mode because this set consists of diverse protein–compound pairs with experimentally verified activities that are appropriate for testing ligand similarity-based screening.

2.3.1. Compatibility Mode: Screening by Binding Site Compatibility Prediction. The compatibility mode of GalaxySagittarius employs a random forest model, which assesses the likelihood of a given protein binding site to be a target site for the query ligand. Since the choice of input features is crucial for a successful model, 47 features for protein binding sites and 17 features for compounds were carefully selected after trial and error, considering the physicochemical principles underlying protein–ligand interactions. Geometrical features such as volume or globularity of ligands are not trivial to define because multiple ligand conformations are possible due to ligand flexibility. It is hoped that features like the number of ligand atoms and the number of rotatable bonds, which are indirectly linked to the shape and size of the ligand, account for geometrical compatibility to the binding site in this low-resolution screening method. When calculating the protein binding site features, amino acid residues within 12.5 Å from the binding site centroid were considered. The selected features are listed in Table 2.

Table 2. Features Used for Binding Site Compatibility Prediction, with the Size of Each Feature Shown in Parenthesis

protein binding site features	compound features
	predicted log P (1)
surface area (1)	number of H-bond donors and acceptors (2)
volume (1)	number of atoms (1)
binding site amino acid composition (20)	number of rotatable bonds (1)
binding surface amino acid composition (20)	topological polar surface area (1)
number of residues (1)	molar refractivity (1)
composition of C, N, O, and S atoms (4)	composition of atom elements (C, N, O, S, P, F, Cl, Br, I, H) (10)

For training and validating binding site compatibility prediction, protein–ligand complexes of the refined set of the PDBbind 2018 database (4463 complexes)²³ were employed. To exclude redundant protein–ligand complex data for unbiased training, the whole data set was first clustered with a protein sequence identity cutoff of 90% using CD-HIT.²⁵ Members of each resulting cluster were clustered again by ligand similarity cutoff of 0.8, where ligand similarity was

measured by the Tanimoto coefficient. The resulting data set comprised 3281 nonredundant protein–ligand complexes.

The nonredundant list of complexes was randomly permuted and separated into a training set (2953 complexes) and a test set (328 complexes). The training set was then split into 5-fold for cross-validation, where 4-fold was used for training and the remaining one was used for validation.

For each compound in the training set, the pool of binding sites to be evaluated by binding site compatibility consisted of 40 randomly selected off-target binding sites in addition to the true binding site. If all off-target sites of the whole training set complexes (2362 sites) are included, a biased learning toward predicting the dominant class (off-targets) can happen. The off-target-to-true-target ratio of 40:1 was chosen after experimenting with various ratios. For validation and tests, the full set of off-targets was used to mimic real situations.

Three different machine learning methods, random forest, extra tree, and extreme gradient boosting, were tested, and a random forest model showed the best performance when measured by the fraction of true binding sites scored within the top 10%. (Performance comparison of the different types of models is provided in the Supporting Information Table S1.) Random forest was therefore selected as the compatibility prediction method of the current prediction pipeline. The random forest model trained on the entire training set is used for the compatibility mode of GalaxySagittarius.

2.3.2. Combined Mode: Screening by Combining Ligand Similarity. It is well known from previous studies that ligand similarity measures can be effective for ligand-based target prediction.^{13–15,26} Therefore, ligand similarity is also used for screening targets in the combined mode of GalaxySagittarius for proteins with available ligands. Similarity of two compounds is measured by the Tanimoto coefficient using the FP2 fingerprint implemented in OpenBabel.²⁷ When a predicted ligand is considered, reliability of the ligand prediction is measured by the structural similarity between the target protein and the homologous protein from which the ligand is taken. The TM-score²⁸ is used as a structural similarity measure, and a TM-score of 1 is assigned if the protein–ligand interaction is obtained from PDB.

The likelihood of the query compound being a binding partner at a given binding site is obtained by a logistic regression using ligand similarity (Tanimoto coefficient), reliability of ligand prediction (TM-score), and the binding site compatibility score (described in the previous subsection) as three variables.

To construct a training set for the regression model, protein–ligand interactions were retrieved from the latest, 25th release of the ChEMBL database²⁴ by the following criteria: (i) interactions involving human protein chains, (ii) interactions annotated as direct binding with the activity measure M (K_i , K_d , IC_{50} , EC_{50}) with $-\log_{10} M \geq 5.5$, (iii) single ligands with <80 heavy atoms, (iv) no contradicting activity reports, and (v) at least one target included in the GalaxySagittarius target database. Redundancy among the compounds satisfying the above criteria was removed by clustering in terms of ligand similarity with the Tanimoto coefficient cutoff of 0.3. The final data set consisted of 3034 compounds with varying numbers of target proteins for each compound.

This situation of multiple possible targets for each compound is different from that of compatibility prediction on the PDBbind set where only a single target corresponds to each compound. Another difference is that the ChEMBL25

database does not provide binding site information, unlike the PDBbind set. Therefore, “predicted” binding sites were used to calculate the binding site compatibility score, while experimentally verified binding sites of the PDBbind set were used for training compatibility prediction. Due to the absence of binding site information in ChEMBL25, any binding site from a known target was considered as a hit when evaluating the prediction results of the combined mode.

Of the 3034 compounds, 2400 compounds were used for developing the predocking score for the combined mode and the remaining 634 compounds were used for testing the performance. The 2400 compounds were again divided into five different groups for cross-validation. The ratio of off-target (inactive interactions) to true target (active interactions) was balanced to 50:1 after trial and error. For the training set of 2400 compounds, the total numbers of active and inactive protein–ligand interactions were 8967 and 423 567 pairs, respectively. The predocking method trained on all entries in the training set is used for the combined mode of GalaxySagittarius.

2.4. Protein–Ligand Docking: Reranking Compounds and Generating Target–Compound Complex Structures.

To fully exploit the available atomic structural details of proteins in the target database, protein–ligand docking is performed against the potential target candidates obtained from predocking calculations. GalaxyDock2,¹⁹ a protein–ligand docking program that employs a powerful global optimization technique called conformational space annealing, is used to search for the optimal ligand binding pose. The GalaxyDock BP2 score²⁹ is used to evaluate the docking poses. Default options are used for docking calculations.

Target candidates are reranked by adding the protein–ligand docking score multiplied by a weight factor w to the predocking score (binding site compatibility score for the compatibility mode and combined score for the combined mode). It was attempted to train the weight parameter w on the set of 634 ChEMBL25 compounds that was also used to test the combined screening method. Among the 634 molecules, the molecules for which targets were successfully found by each predocking method within the top 1000 were used for training w , resulting in 391 molecules for the compatibility mode and 543 molecules for the combined mode. One-fifth of the compounds were selected as a test set, and the rest were used as a training set. The parameter w that maximizes the average rank of the true targets was chosen for each prediction mode of GalaxySagittarius.

Ligand three-dimensional (3D) conformations for docking were generated using the 3D structure generation program CORINA³⁰ from sdf files of ChEMBL25.

3. RESULTS AND DISCUSSION

3.1. Results for Predocking Screening Methods.

Performances of the predocking methods in screening “binding sites” are first discussed for each of the two prediction modes: the compatibility mode (Section 3.1.1) and the combined mode (Section 3.1.2). Performances of selecting 1000 “proteins” by the predocking methods are discussed next (Section 3.1.3).

3.1.1. Compatibility Mode: Evaluating Binding Sites by Compatibility Prediction. Performance of the binding site compatibility prediction as a ranking method for binding sites was measured as the percentage of true targets ranked in the top $N\%$ and that as a discriminator of the true binding sites

was measured by AUC, area under the receiver operating characteristic (ROC) curve. Performance in the 5-fold cross-validation on the training set and that of the final model on the test set is summarized in Table 3. Since prediction perform-

Table 3. Performance of the Binding Site Compatibility Prediction in Terms of the Percentage of the True Binding Sites Ranked in the Top N% and AUC

	top 2%	top 5%	top 10%	AUC
validation	38.1%	55.1%	67.7%	0.887
test	42.1%	58.8%	70.1%	0.905

ance is expected to heavily depend on the characteristics of the data set, the results are not compared to other methods evaluated on different data sets. Still, the relatively high AUC values of 0.887 and 0.905 for the validation set and the test set, respectively, and close to 70% of the true binding sites ranked within the top 10% seemed reasonable for a principle-based prescreening method.

The compatibility prediction method was designed to incorporate physicochemical principles underlying protein–ligand interactions. Analysis of feature importance of the prediction model shows that features directly related to binding, such as amino acid composition of the binding site and element composition and hydrophobicity of the ligand, are more important for compatibility than general molecular features such as the number of residues of protein. (See the Supporting Information Text S1 and Figure S1 for more details.) Although the binding site compatibility prediction is not as physically thorough as protein–ligand docking, which directly estimates protein–ligand interaction free energy, it is much more computationally efficient than docking and thus adequate as a screening method.

3.1.2. Combined Mode: Evaluating Binding Sites by Combining Ligand Similarity. While the compatibility mode focuses on finding new classes of targets based on physicochemical principles, the combined mode combines the available information of binding ligands, assuming that an information-based prediction tends to be more accurate when relevant information is available. The ligand similarity element of GalaxySagittarius has an additional advantage of the enriched database of protein–ligand interactions. The number of protein–ligand interactions increased from 45 416 to 389 237 by adding interactions predicted by GalaxySite²⁰ to those collected from PDB. The added interactions can boost the target prediction performance, as discussed later (Section 3.2.1).

Effectiveness of the combined prescreening method as a discriminator of the true binding sites was measured by AUC of the ROC curves for the 5-fold cross-validation results on the training set of 2400 ChEMBL25 compounds, as shown in Figure 2. ROC curves for the prediction by ligand similarity only and those for the prediction by binding site compatibility only are compared together in the figure. The average AUC value for the combined prediction (0.797) is slightly higher than that by ligand similarity only (0.781), while that by binding site compatibility is much lower (0.680). [This AUC value for compatibility prediction is much lower than those on the PDBbind set in Table 3 (0.885 and 0.874) because predicted binding sites are used here for the ChEMBL25 set, while “PDB” binding sites are used for the PDBbind set.] This result implies that the ligand similarity component is the major

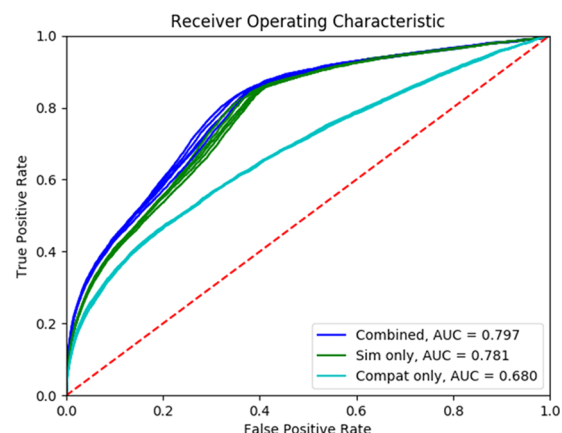


Figure 2. ROC curves comparing the powers of discriminating active from inactive interactions by the combined method (colored in blue), by ligand similarity only (green), and by compatibility prediction only (cyan).

contributor of the predocking score in the combined mode. Noting that similar ligands with a Tanimoto coefficient > 0.3 were removed in the ChEMBL25 set, the above results illustrate proved successful in detecting lower level ligand similarities.

The combined predocking method of GalaxySagittarius on the ChEMBL25 set is compared to results of other programs reported in ref 18 in Table 4. This set of AUC values cannot be interpreted as a strict performance comparison of different programs because they were tested on different sets. It can still be concluded with caution that performance of the combined predocking method of GalaxySagittarius is comparable to those of other available methods, especially considering the large size of the current set.

3.1.3. Performance of Selecting 1000 Candidates by Predocking Screening. The performance of predocking screening methods in selecting 1000 targets from the GalaxySagittarius database was evaluated on a test set of 634 ChEMBL25 compounds that are involved in 1300 protein–ligand interactions with the average/median number of interactions per compound of 2.05/1. Table 5 shows the percentage of known protein targets in the 1000 candidates selected from the database [from 11 768 (8279) proteins and protein interfaces in the compatibility (combined) mode]. Success rates of ranking targets within the top 2, 5, and 10% of the database are shown together. The combined method shows the highest success rate of 78.2% for the top 1000 selection, compared to 73.6 and 57.4% for the methods based on ligand similarity only and that based on binding site compatibility only, respectively. The success rate of the compatibility mode for the top 1000 selection is lower than that at 10% because the top 10% corresponds to 1177 proteins (the target database size is 11 768). Random selection of 1000 proteins would give success rates of 28.9 and 16.5% for the combined and the compatibility database, respectively.

It can be assumed that the current test set derived from ChEMBL25 tends to be biased toward previously studied targets by the community. We still had to use this set due to the need for having experimentally verified active targets for model validation. Although the apparent performance of the compatibility mode on the ChEMBL25 set is lower than that of the combined mode, the compatibility mode still has an advantage of searching the target space not much explored

Table 4. Comparison of the GalaxySagittarius Predocking Method of the Combined Mode on the ChEMBL25 Set to Other Target Prediction Programs on Various Test Sets

	method	set	prediction performance
		active pair/inactive pair	AUC
ChemMapper	ligand 3D similarity	216/7069	0.7
ChemProt 3.0	ligand 2D similarity	248/1700	0.827
SwissTargetPrediction	2D + 3D similarity	346/1730	0.87
idTarget	docking	3/1161 and 4/1161	0.89 and 0.91
ACID	docking	133/831	0.84
GalaxySagittarius combined predocking	similarity and compatibility	8967/423 567	0.80

Table 5. Predocking Results of the Combined Mode of GalaxySagittarius in Terms of the Percentage of True Targets Found in the Top 1000 and in the Top 2, 5, and 10% When Tested on 634 ChEMBL25 Compounds

	percentage of true targets ranked in	
	top 1000	top 2/5/10%
combined	78.2	46.2/61.5/74.3
similarity	73.6	44.8/60.0/70.8
compatibility	57.4	36.6/48.9/59.5

previously. In fact, the similarity-based method and the compatibility-based method are highly complementary with each other, sharing only 30.4% of the top 1000 candidates in the current test.

Moreover, the compatibility method could identify targets that could not be detected by combining ligand similarity because the method can evaluate proteins without available ligand information. For example, a known target of ChEMBL ID: 275938, DNA-3-methyladenine glycosylase (Uniprot ID: P29372), was ranked 76th by the compatibility method, but the target was not even considered in the combined mode because it lacked a binding ligand in the database. Similarly, for ruxolitinib (ChEMBL ID: 1789941), the first FDA-approved drug for treatment of myelofibrosis,³¹ a protein target MAP3K3 (mitogen-activated protein kinase 3; Uniprot ID: Q99759) without ligand interaction in the database was detected by compatibility prediction. The compatibility mode of GalaxySagittarius was able to retrieve 67 of 75 known targets of ruxolitinib within the top 1000 among which the protein with the highest known affinity ($pK_d = 10.44$), JAK2 (Uniprot ID: O60674), ranked first and MAP3K3 with a lower affinity ($pK_d = 6.82$) ranked 908th.

This result implies that the compatibility method of GalaxySagittarius has the potential to detect new classes of potential targets that may not be possible with other similarity-based target prediction methods.

3.2. Performance of GalaxySagittarius after Combining Protein–Ligand Docking. **3.2.1. Target Reranking Results Using the Protein–Ligand Docking Results.** It is of interest whether protein–ligand docking methods can assist in a large-scale target search, with various technical hurdles such as requiring accurate protein structures and binding sites. In the case of GalaxySagittarius, docking could improve target ranking when combined with predocking scores, as discussed below. Moreover, docking has a great advantage of providing insights into atomic details of protein–ligand interactions, which can be further utilized in molecular designs.

The reranking results of the final score that combines the predocking score and the docking score (estimated binding free energy) on the training and test sets are compared to those

of the predocking score only and the docking score only in Table 6 for the compatibility mode and in Table 7 for the

Table 6. Comparison of Results by the Final Reranking Score to Those by the Predocking Score and by the Docking Score for the Compatibility Mode of GalaxySagittarius

	results for the compatibility mode			
	top 100/top 200 success rate ^a (%)		average rank of true targets (%)	
	training set	test set	training set	test set
final	44.4/58.7	50.7/66.9	74.0	73.2
predocking	44.5/56.2	47.6/60.7	69.5	69.2
docking	25.7/42.5	24.8/42.8	68.3	66.5

^aPercentage of known targets identified within the top 100/top 200 among 1000 candidates by each score.

Table 7. Comparison of Results by the Final Reranking Score to Those by the Predocking Score and by the Docking Score for the Combined Mode of GalaxySagittarius

	results for the combined mode			
	top 100/top 200 success rate ^a (%)		average rank of true targets (%)	
	training set	test set	training set	test set
final	48.1/58.5	47.5/59.4	71.7	76.9
predocking	47.5/58.6	48.9/58.7	70.3	75.7
docking	19.3/34.3	19.7/31.4	60.5	63.8

^aPercentage of known targets identified within the top 100/top 200 among 1000 candidates by each score.

combined mode. The average rank in the test set improved from 69.2% by the predocking score and 66.5% by the docking score to 73.2% by the combined score in the compatibility mode and from 75.7 and 63.8 to 76.9% in the combined mode. Overall trends are the same on the training set, indicating no overtraining. The small contribution of docking to reranking in the combined mode implies current limitation in docking, but docking can still help improve ranking, especially in the compatibility mode. The larger contribution of docking in the compatibility mode than in the combined mode seems to originate from the fact that the potential targets selected by binding site compatibility prediction have better chances of having high docking scores.

The percentage of known targets scored within the top 100 among 1000 candidates (top 100 success rate) increased from 47.6% of the predocking score to 50.7% by combining the docking score in the compatibility mode, although the success rate is only 24.8% by the docking score. The top 100 success rate slightly decreased from 48.9% of the predocking score to 47.5% by combining the docking score in the combined mode.

3.2.2. Predicting the Target–Compound Complex Structure by Protein–Ligand Docking. The target–compound complex structure is expected to be predicted accurately when an appropriate target protein structure is available in the database. An example is lapatinib (ChEMBL ID: 554), a drug for breast cancer and other solid tumors, which mainly works as a dual inhibitor of epidermal growth factor receptor (EGFR) and human epidermal growth factor receptor 2 (HER2) tyrosine kinases.³² EGFR (Uniprot ID: P00533) was ranked first by both modes of GalaxySagittarius along with other detected known targets. The binding site with the highest binding site compatibility score for EGFR was that from PDB ID 3EBL, which is a structure bound to another ligand called POX with a Tanimoto coefficient of 0.324 from lapatinib. The predicted EGFR–lapatinib complex structure by protein–ligand docking is very close to the crystal complex structure of the EGFR–lapatinib complex (PDB ID: 1XKK) with ligand root-mean-square deviation (RMSD) of 1.48 Å, as shown in Figure 3. As for the other main target of lapatinib, HER2

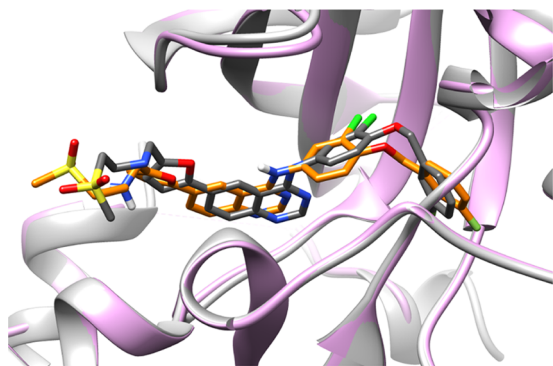


Figure 3. Complex structure of the query compound lapatinib with a predicted target EGFR obtained by docking lapatinib to the EGFR binding site detected by compatibility prediction (PDB ID: 3EBL). The predicted complex structure (orange ligand structure and pink protein structure) is very close to the crystal complex structure with the ligand RMSD = 1.48 Å (gray, PDB ID: 1XKK).

protein, there is no known cocrystal structure available for comparison, but GalaxySagittarius provides a predicted complex structure based on docking. Among other proteins known to bind lapatinib, 16 and 19 different targets were predicted within the top 1000 by the compatibility mode and the combined mode, respectively.

3.2.3. Benefits of Using Predicted Binding Sites and Ligands. The target database of GalaxySagittarius includes predicted binding ligands and binding sites, resulting in increase in the database size, as shown in Table 1. For example, 343 821 new protein–ligand interactions including 1007 new ligands for which no bound poses were reported previously on human proteins were included. The effect of this enlarged database was examined for the test cases of 634 ChEMBL25 compounds. Predicted binding information covered 116 of 1300 known protein–ligand interaction pairs for the 634 compounds, meaning that the 116 interactions would not have been considered without the predicted information. Among the 116 interactions, 14 (11) interactions for 14 (11) different compounds were detected within the top 1000 by the combined (compatibility) mode. Two (two) target predictions of 14 (11) by the combined (compatibility)

mode were for compounds for which no single targets would have been found without predicted binding information.

For example, one of the targets of brinzolamide (ChEMBL ID: 220491), an FDA-approved drug inhibiting carbonic anhydrase (CA),³³ CA6 (Uniprot ID: P23280), could not have been identified as a target if predicted binding information were not included. The binding site of CA6 was predicted with several ligands, including a ligand ETS with a high Tanimoto coefficient of 0.496 from brinzolamide. CA6 was ranked 17th in predocking and 11th by reranking after docking by the combined mode.

3.3. GalaxySagittarius Web Server. The GalaxySagittarius server (<http://galaxy.seoklab.org/sagittarius>) runs on a cluster of Linux servers, which utilizes 252 2.40 GHz Intel Xeon processors. Users have to submit the input compound as a SMILES string or a structure file (PDB, MOL2, or XYZ format). The average run time of GalaxySagittarius is about 6 h. Since most of the time is consumed to perform protein–ligand docking and the performance is marginally improved by adding docking in the combined mode (Table 7), a fast prediction mode that does not run docking is also provided. The fast mode takes about 10 min. Details of the selected top 200 proteins can be examined on the web, and the predictions and binding poses can be downloaded.

4. CONCLUSIONS

A new target prediction method called GalaxySagittarius was developed by combining structure- and similarity-based approaches. In the structure-based, compatibility mode of GalaxySagittarius, an efficient binding site compatibility prediction was applied for fast and accurate screening of protein binding sites to select promising binding sites for docking. This prediction mode based on the binding principle could select targets without any binding partners deposited in the structure database in test cases. In the combined mode of GalaxySagittarius, a screening score that combines ligand similarity was employed before docking for accurate and reliable prediction when ligand information is available in the database. Both modes of GalaxySagittarius are benefited by the enriched database with predicted binding ligands and sites, as illustrated by several test cases in which targets without cocrystallized ligands were identified based on the predicted ligands and/or sites added to the database. The protein–ligand docking stage of GalaxySagittarius was able to provide docking poses with correct atomic details when appropriate protein structures and sites were available in the database.

While the current method seems to be at least comparable to previously developed target prediction methods, there is still a large room for improvement in absolute prediction accuracy. The average percentage of selecting known targets in 1000 is near 80%, and ranking the known targets in 100 among 1000 is close to 50%, when tested on a compiled ChEMBL25 data set. The probability of success in actual applications can heavily depend on the nature of the problem at hand. We still expect that the current method can be useful especially because it is complementary to other methods due to the compatibility prediction and predicted ligand information that are unique to the current method. With the rapidly increasing structure database and binding information, this kind of tool would become a more promising tool for target prediction in the future.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.0c00104>.

Analysis of feature importance in the random forest model for binding site compatibility prediction; performance comparison of XGBoost (extreme gradient boosting), extra tree, and random forest models in binding site compatibility prediction (PDF)

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Notes

The authors declare no competing financial interest.

A web server is freely available at <http://galaxy.seoklab.org/sagittarius>.

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